

LAB 08

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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score

df = pd.read_csv('Breast Cancer Dataset.csv')

X = df.drop(['id', 'diagnosis'], axis=1)
y = df['diagnosis']

y = y.map({'M': 1, 'B': 0})

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

clf = DecisionTreeClassifier(max_depth=3, random_state=42)
clf.fit(X_train, y_train)

y_pred = clf.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print(f"Model Accuracy: {accuracy * 100:.2f}%")
```

```
new_sample = X_test.iloc[[0]]  
prediction = clf.predict(new_sample)
```

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prediction_class = "Malignant" if prediction[0] == 1 else "Benign"  
print(f"New Sample Classification Result: {prediction_class}")
```

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plt.figure(figsize=(20, 10))  
plot_tree(  
    clf,  
    feature_names=X.columns,  
    class_names=['Benign', 'Malignant'],  
    filled=True,  
    rounded=True,  
    fontsize=12  
)  
plt.title("Decision Tree Visualization (Breast Cancer Data)")  
plt.show()
```